$(1056)_{10} \xrightarrow{\text{conversione}}$

- $?_2$
- $?_8$
- $?_{16}$

1056	0
528	0
264	0
132	0
66	0
33	1
16	0
8	0
4	0
2	0
1	1

$\longrightarrow$

10000100000 Base 2

$\underbrace{10}_{2}\underbrace{000}_{0}\underbrace{100}_{4}\underbrace{000}_{0}$ Base 8 Gruppi da 3 bit --> 2^3

$\underbrace{100}_{4}\underbrace{0010}_{2}\underbrace{0000}_{0}$ Base 16 Gruppi da 4 bit --> 2^4

↘

Divisione in colonna per ottenere la base 2
Se divisibile = 0
Se non é divisibile = 1 [rip x difetto]

$(1F7)_{16} \xrightarrow{\text{conversione}}$

- $?_2$
- $?_8$
- $?_{10}$

1	F	7
0001	1111	0111

$\longrightarrow$ 111110111 base 2

$\underbrace{111}_{7}\underbrace{110}_{6}\underbrace{111}_{7}$ base 8

$(64)_8 \xrightarrow{\text{conversione}}$

- $?_2$
- $?_{10}$
- $?_{16}$

6	4
110	100

$\longrightarrow$ 110100 base 8

$\underbrace{11}_{3}\underbrace{0100}_{4}$ base 16

$\underbrace{3 \quad 4}$

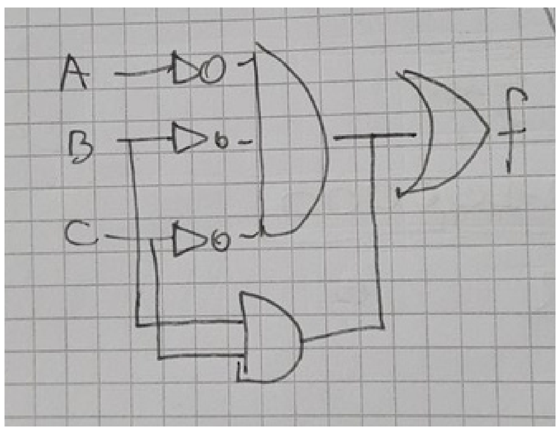
$(34)_H$

Metodo mappe K

$$A + \bar{A}B = A + B$$

$$\bar{a} \cdot \bar{b} \cdot \bar{c} + \bar{a} \cdot bc + abc$$

a/bc	00	01	11	10
0	1		1	
1			1	



$$\overline{abc} + bc$$

a	b	c	f	
0	0	0	1	$\longrightarrow$ 0
0	0	1	0	
0	1	0	0	
0	1	1	1	$\longrightarrow$ 3
1	0	0	0	
1	0	1	0	
1	1	0	0	
1	1	1	1	$\longrightarrow$ 7

$$\Sigma\,(0, 3, 7)$$

a/bc	00	01	11	10
0	1		1	
1			1	

$$\overline{abc} + \bar{a}bc + abc$$

$$a\bar{b} + \bar{a} \cdot \overline{a\bar{b}} + c\bar{d} + cd$$

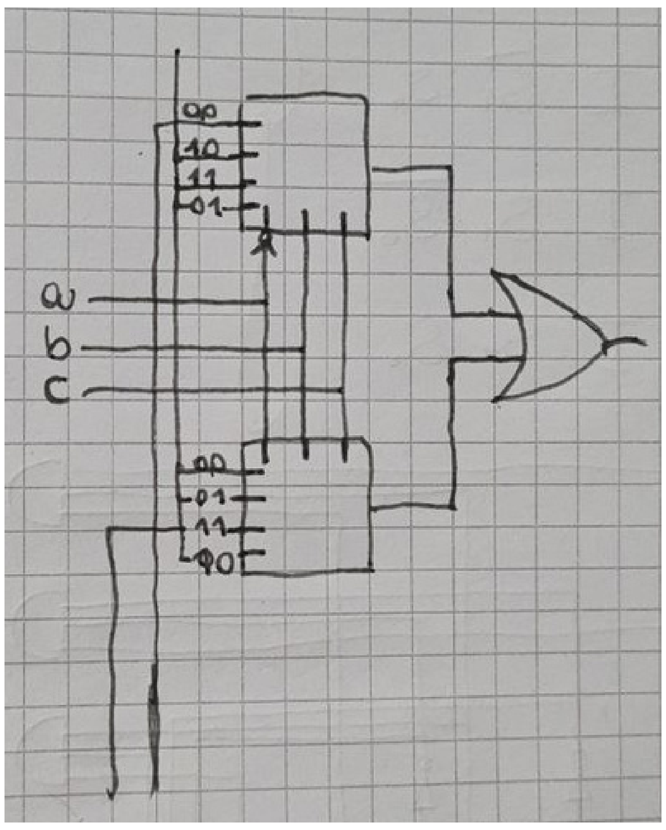
step 1 : Trovare la forma minima

$$= a\bar{b} + \bar{a}(\bar{a} \cdot b) + c(d + \bar{d})$$

$$= a\bar{b} + \bar{a}b + c \longrightarrow \text{Forma minima}$$

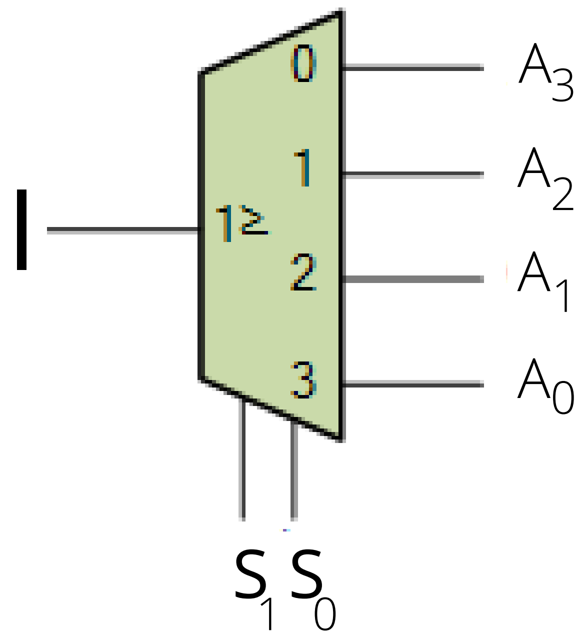
a	b	c	f
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

step 2
Studio della
tavola di
veritá



Step 3:
Studio del circuito

DEMULTIPLEXER



$f(I\ S_1, S_0)$

$f(I\ S_1, S_0)\ \{0,1\}$

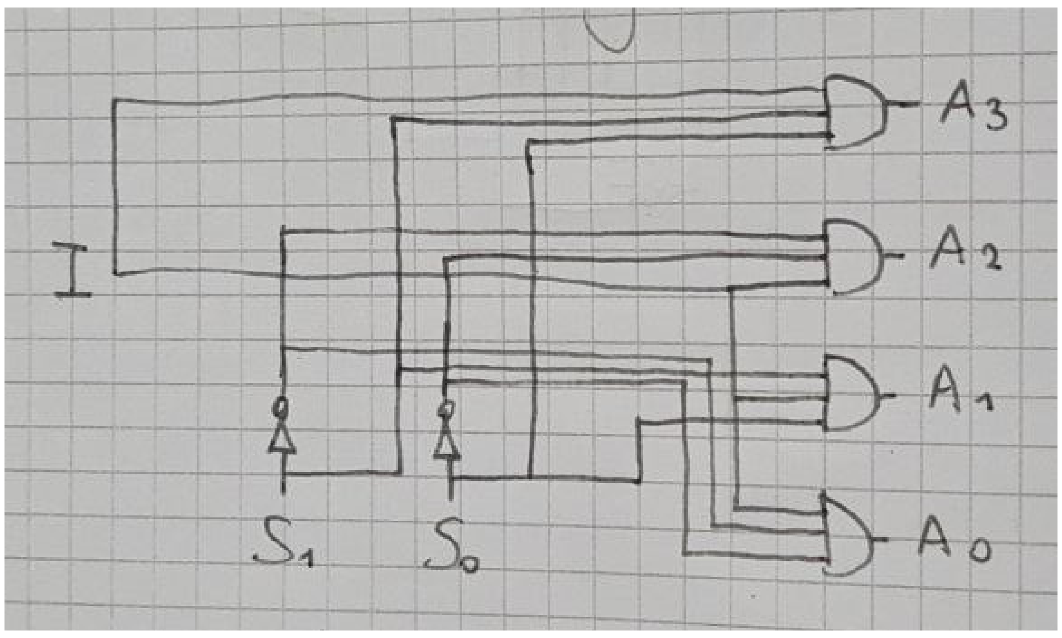
I	S_1	S_0	A_0	A_1	A_2	A_3
0	0	0				
0	0	1				
0	1	0				
0	1	1				
1	0	0	1	0	0	0
1	0	1	0	1	0	0
1	1	0	0	0	1	0
1	1	1	0	0	0	1

$A_0 = I \cdot \overline{S_1} \cdot \overline{S_0}$

$A_1 = I \cdot \overline{S_1} \cdot S_0$

$A_2 = I \cdot S_1 \cdot \overline{S_0}$

$A_3 = I \cdot S_1 \cdot S_0$



Codice di correzione Hamming

$F5_H \xrightarrow{\text{conversione}} 11110101$

$m + n + 1 <= 2^n$

$9 + n <= 2^n$

numero bit aggiuntivi

In XOR

Se gli 1 sono pari = 0
Se gli 1 sono dispari = 1